

FUNCTIONS

1. Add Two Numbers

Write a function that takes two numbers as input and returns their sum.

2. Find Maximum of Two Numbers

Write a function that accepts two integers and returns the greater number.

3. Check Even or Odd

Write a function that takes an integer and returns `true` if it's even, otherwise `false`.

4. Print Multiplication Table

Write a function that takes a number `n` and prints its multiplication table up to 10.

5. Unique Digit Sum Checker

Write a function `hasUniqueDigitSum(int num1, int num2)` that:

- Calculates the sum of digits of both numbers.
- Returns `true` if the digit sums are not equal, `false` otherwise.

6. Password Strength Evaluator

Write a function `isStrongPassword(String password)` that returns `true` if:

- It is at least 8 characters long
- Contains at least one digit, one uppercase, one lowercase, and one special character

7. Compress String

Write a function `compress(String input)` that converts `"aaabbcdd"` → `"a3b2c1d2"`

(Counts consecutive repeating characters.) **8.**

Find First Non-Repeating Character

Problem:

Write a function `firstUniqueChar(String s)` that returns the first non-repeating character in a string.

FUNCTION OVERLOADING

1. Area Calculator (Overload based on shape)

Write a class `AreaCalculator` with overloaded `area()` methods to compute the area of:

- a circle: `area(double radius)`
- a rectangle: `area(double length, double breadth)`
- a triangle: `area(double base, double height)`

2. Print

Utility (Overload for different data types)

Create a class `Printer` with an overloaded `print()` method:

- `print(int value)`
- `print(String message)`
- `print(double value)`
- `print(char c)`

3. Calculator with Optional Power (Varying Parameters)

Create a class `Calculator` with an overloaded method `calculate(int base)` and `calculate(int base, int power)`.

- First version returns `base * base`
- Second version returns `base ^ power`

4. Sum of Numbers (Variable-length arguments)

Create a method `sum(int a, int b)` and another method `sum(int... numbers)`.

- The first sums two numbers.
- The second sums an arbitrary number of integers.

Test case: `sum(2,3)` and `sum(1,2,3,4,5,6)`.

5. Booking System (Different Booking Types)

Create a class `BookingSystem` with multiple `book()` methods:

- `book(String name)` – for normal booking
- `book(String name, int members)` – for group booking
- `book(String name, boolean isVIP)` – for VIP booking
- `book(String name, int members, boolean isVIP)` – group VIP booking